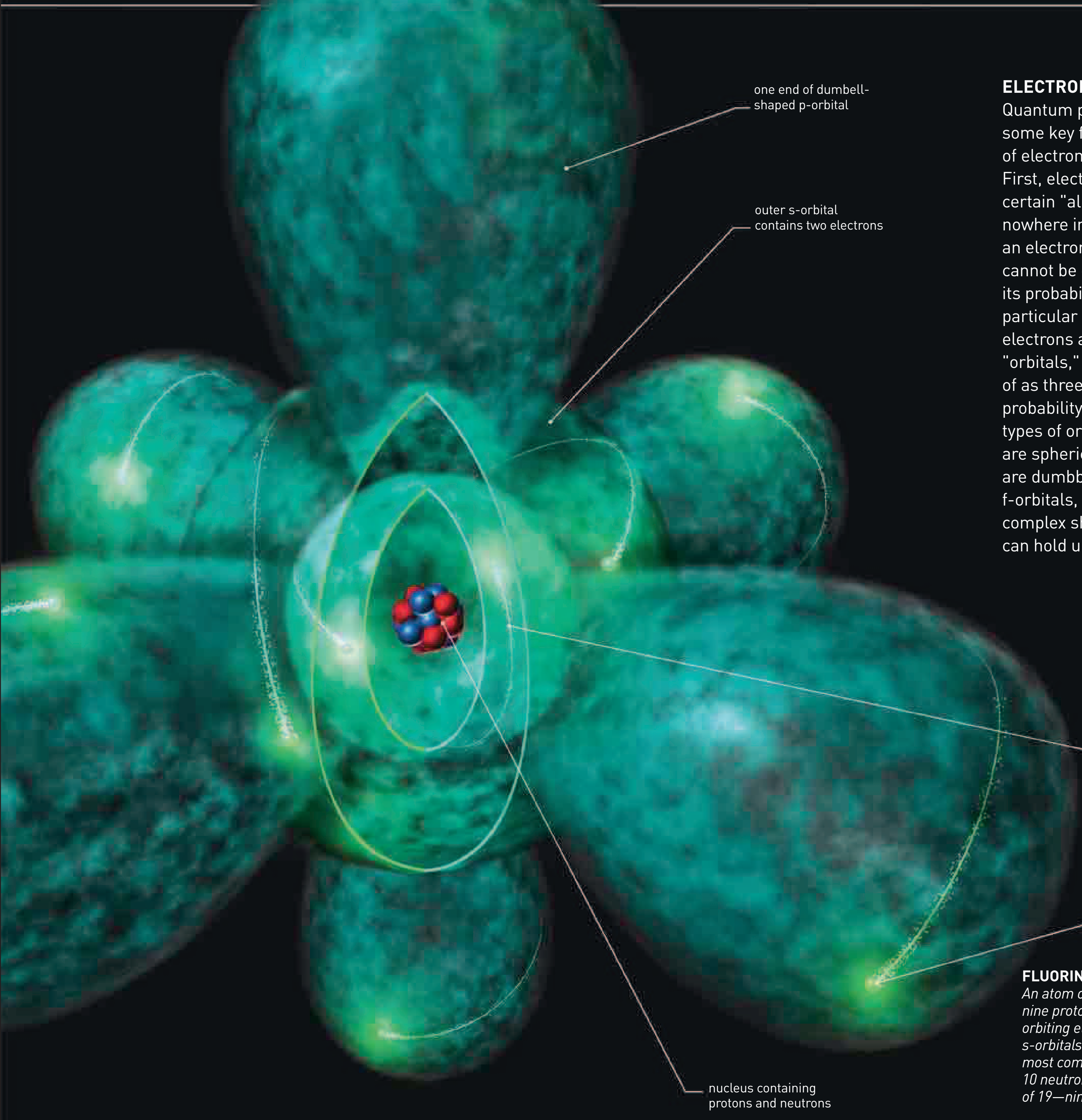

ELECTRON ORBITALS

Quantum physics has revealed some key facts about the orbits of electrons around the nucleus. First, electrons can only occupy certain "allowed" orbits, and nowhere in between. Second, an electron's exact position cannot be determined, only its probability of existing in a particular place. As a result, electrons actually exist in "orbitals," which can be thought of as three-dimensional clouds of probability. There are four main types of orbital: s-orbitals, which are spherical; p-orbitals, which are dumbbell-shaped; and d- and f-orbitals, which have more complex shapes. Each orbital can hold up to two electrons.

FLUORINE ATOM
An atom of the element fluorine has nine protons in its nucleus and nine orbiting electrons arranged in two s-orbitals and three p-orbitals. The most common version of fluorine has 10 neutrons, so it has an atomic mass of 19—nine protons plus 10 neutrons.

INCANDESCENCE AND LUMINESCENCE

There are two ways in which matter emits light: incandescence and luminescence. Incandescence is the production of light from hot matter—things glow red hot or white hot, for example. In contrast, luminescence does not require heat. Fluorescence, glowing when illuminated by ultraviolet radiation, and phosphorescence, glowing in the dark after being illuminated, are two familiar examples. Light produced by luminescence is the result of energy lost by electrons "falling" to a lower energy level, closer to an atom's nucleus.